FINAL YEAR PROJECT · 2026

A.U.R.A.

A Locally-Run Multi-Agent System for Secure Computer & OS Automation

Air-gapped · Unified · Responsive · Agent

LOCAL | MULTI-AGENT | ZERO-TRUST



TEAM

Muhammad Hunzala Hasan — 25435

Muhammad Taha Jabbar — 25421

Hafsa Ayub — 25588

ADVANCED · UNIFIED · RESPONSIVE · ASSISTANT

A Multi-Domain System

A.U.R.A. sits at the intersection of four engineering domains — each one is a constraint the system must respect.

ARTIFICIAL INTELLIGENCE

- Local LLMs & multi-agent reasoning
- Task interpretation · Planning & decomposition
- Local model inference · Agent coordination

COMPUTER AUTOMATION

- OS & application interaction
- Accessibility-based UI targeting
- Background task execution · File & app operations
- Human-like interaction fallback

CYBERSECURITY

- Privacy, access control & agent safety
- Zero-trust execution · Prompt-injection defense
- Restricted permissions · Human approval for high-risk actions

EDGE COMPUTING

- Offline & hardware-assisted computing
- Local inference · CPU-efficient quantized models
- Raspberry Pi companion · Local encrypted state

Useful automation requires all four — a single-domain solution is not enough.

Why Existing AI Assistants Fall Short

Five gaps between how today's assistants are designed and what local computer automation actually requires.

CLOUD DEPENDENCY

Reasoning and user context leave the machine.

LIMITED OS CONTROL

Cannot reliably perform multi-step desktop workflows.

WEAK CONTEXT PERSISTENCE

Preferences fragment across sessions.

SECURITY RISKS

Tool-using agents face malicious content and over-broad permissions.

NO SAFE APPROVAL LAYER

A confirm dialog is weaker than a meaningful authorization gate.

Existing assistants treat capability and control as a trade-off — usually defaulting to capability.

Three Pressures on the Modern Computer User

REPETITIVE TASKS

- Open apps
- Manage files
- Run searches
- Move between windows
- Repeat workflows

SENSITIVE DATA

- Personal documents
- Credentials
- Work files
- Private conversations
- Browser activity

TRUST & CONTROL

- Users want automation
- without giving an AI
- unrestricted system access.

"Can an AI agent perform useful computer tasks while keeping the user's data, reasoning and execution locally controlled?"

— A.U.R.A. is designed around this question. —

Conventional Assistant → A.U.R.A.

CONVENTIONAL AI ASSISTANT		A.U.R.A.
Cloud-based processing	→	**Local processing**
API-dependent intelligence	→	**Local quantized models**
Chat-oriented interaction	→	**Task-oriented OS automation**
Single assistant workflow	→	**Multi-agent orchestration**
Limited persistent context	→	**Hierarchical long-term memory**
Accessibility often secondary	→	**Accessibility-first interaction**
Vision as primary method	→	**Vision as fallback**
Broad permissions (risk)	→	**Restricted-permission execution**
Basic confirmation dialogs	→	**Biometric HITL approval**
Sensitive state may leave device	→	**Encrypted local storage**

Useful automation without sacrificing local privacy and user control.

Six Stages From Command to Result

A single end-to-end view — the rest of the deck unpacks each stage.

01

USER COMMAND

Natural-language via local UI or voice.

02

BRAIN

Interprets intent and identifies the task.

03

MANAGER

Decomposes the task; selects sub-agents.

04

SUB-AGENTS

Browser · File/OS · Comm · Security · Research.

05

LOCAL EXECUTION

Accessibility APIs, approved tools, sandboxing.

06

VERIFY & RESPOND

Validate, record state, return outcome.

DATA SPINE · shared context flows across all six stages

Brain interprets. Manager plans. Sub-agents execute under rules. Manager verifies.

Ten Engineered Capabilities

Each capability corresponds to a concrete subsystem. None are aspirational.

01

MULTI-AGENT ORCHESTRATION

Brain, Manager, and specialized agents cooperate with shared context.

02

HYBRID LOCAL REASONING

BitNet b1.58 2B for routing + quantized 7–8B for complex reasoning.

03

BACKGROUND OS AUTOMATION

Accessibility APIs target UI elements directly.

04

VISION FALLBACK

Screenshot-based understanding when accessibility metadata is missing.

05

LONG-TERM MEMORY

Habits, preferences, and references for cross-session continuity.

06

ZERO-TRUST SECURITY

Untrusted web content is sanitized before reaching the AI.

07

CONTROLLED INTERACTION

Platform-aware rate limits and interaction policies.

08

BIOMETRIC HITL

High-risk actions pause for fingerprint verification.

09

RASPBERRY PI COMPANION

Wake-word mic · fingerprint sensor · projector feedback.

10

ENCRYPTED LOCAL STORAGE

AES-256 protects memory and local state databases.

From Architecture to a Real Device



FUNCTIONAL LAYERS

12 components, 4 roles.

SENSING

- (6) Mic Array 6-MIC
- (3) Fingerprint Sensor
- (2) 5MP Camera OV5647
- (12) Ambient Light Strip

COMPUTE

- (5) Raspberry Pi 4 Model B
- (11) USB Hub 4-port
- (4) Custom I/O & Sensor Board

REASONING + UI

- (1) 45° Holographic Display
- (9) Speaker 5W
- (10) LED Status Indicators

POWER + PROTECTION

DIMENSIONS 150 × 150 × 135 mm · WEIGHT ~1.2 kg · CONNECTIVITY Wi-Fi · Bluetooth · USB 2.0/3.0 · STORAGE 32 GB MicroSD · OS Custom Secure Linux

Multi-Agent Task Execution

One example task, six stages, five specialized agents.

EXAMPLE TASK

"Find the project report, summarize it, and save the summary in the Documents folder."

STEP 01 — **USER INPUT**
Natural-language command issued by the user.

↓

STEP 02 — **INTENT UNDERSTANDING (Brain)**
Identifies: user goal · required apps · required info · expected result.

↓

STEP 03 — **TASK DECOMPOSITION (Manager)**
Locate report · Read content · Generate summary · Open target folder · Save output · Verify completion.

↓

STEP 04 — **AGENT ROUTING**
File/OS · Research · Browser · Communication · Security.

↓

STEP 05 — **CONTROLLED EXECUTION**
Accessibility API → element found? → YES: direct / NO: vision fallback.

↓

STEP 06 — **RESULT VERIFICATION**

AGENT ROUTING → STEP 04

Manager dispatches to the right specialist.

Browser Agent	Locate report on the web.
File / OS Agent	Open target folders, save outputs.
Research Agent	Pull relevant content snippets.
Communication Agent	Email / message follow-ups.
Security Agent	Validate permissions, gates.

STEP 05 · Accessibility API → UI element found? → YES: Direct interaction | NO: Vision fallback

Hybrid Quantized Local Reasoning

Two models on the same device. Routing is itself a model decision.

ONE DEVICE · TWO TIERS

ROUTING / LIGHTWEIGHT

BitNet b1.58 2B

Fast classification · intent routing · simple decisions · low-resource inference.

- Classify task type
- Route to the right sub-agent
- Confirm whether vision fallback is needed
- Detect language / domain of the user input

ROUTE

Simple task → lightweight

Complex task → larger model

COMPLEX REASONING

Quantized 7–8B LM

Multi-step reasoning · task planning · context interpretation · complex decisions.

- Decompose multi-step commands
- Generate the summary content
- Plan file/folder operations end-to-end
- Resolve ambiguous or partial user requests

WHY HYBRID?

Running the larger model for every operation is wasteful. Simple tasks route to BitNet; complex reasoning routes to the 7–8B model.

- *Expected benefits: reduced compute overhead · better CPU use · lower local inference cost · more practical offline operation.*

Memory, Security & Agent Control

Four supporting modules that turn a multi-agent loop into a trustworthy loop.

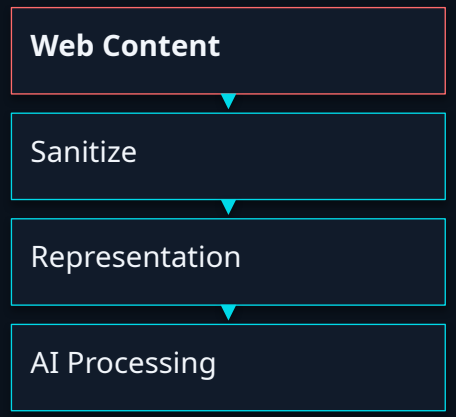
01 · LONG-TERM MEMORY

- Preferences
- Repeated Workflows
- Useful File Paths
- Historical Context

ChromaDB · Local retrieval.

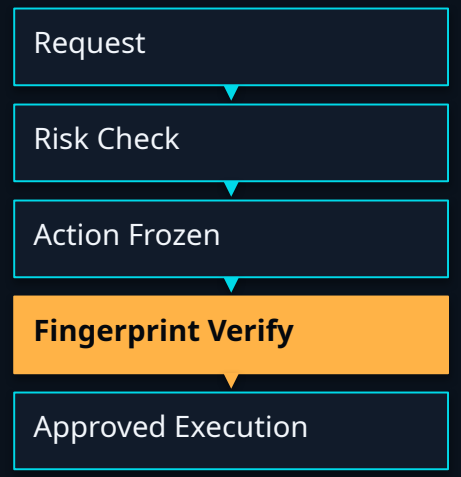
Stores habits, preferences, and references so the system is the same person across sessions.

02 · SECURITY LAYER



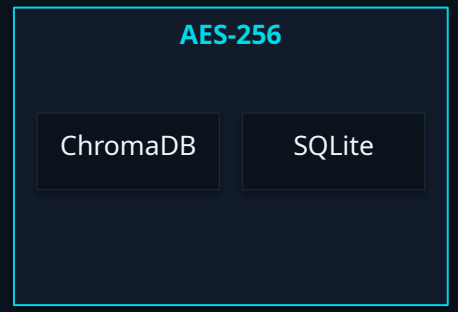
External content never reaches the AI in raw form. Reduces indirect prompt-injection surface.

03 · BIOMETRIC CONTROL



Physical biometric pause before high-risk actions. The fingerprint ring is amber — that is the user's only cue.

04 · ENCRYPTED STATE



Local-only state management. No remote sync.

Memory and state live on the same encrypted volume. Loss of the device is loss of encryption keys, not data leakage.

From User Command to Verified Result

One flowchart that the entire system follows. Decisions are diamonds; gates are amber.

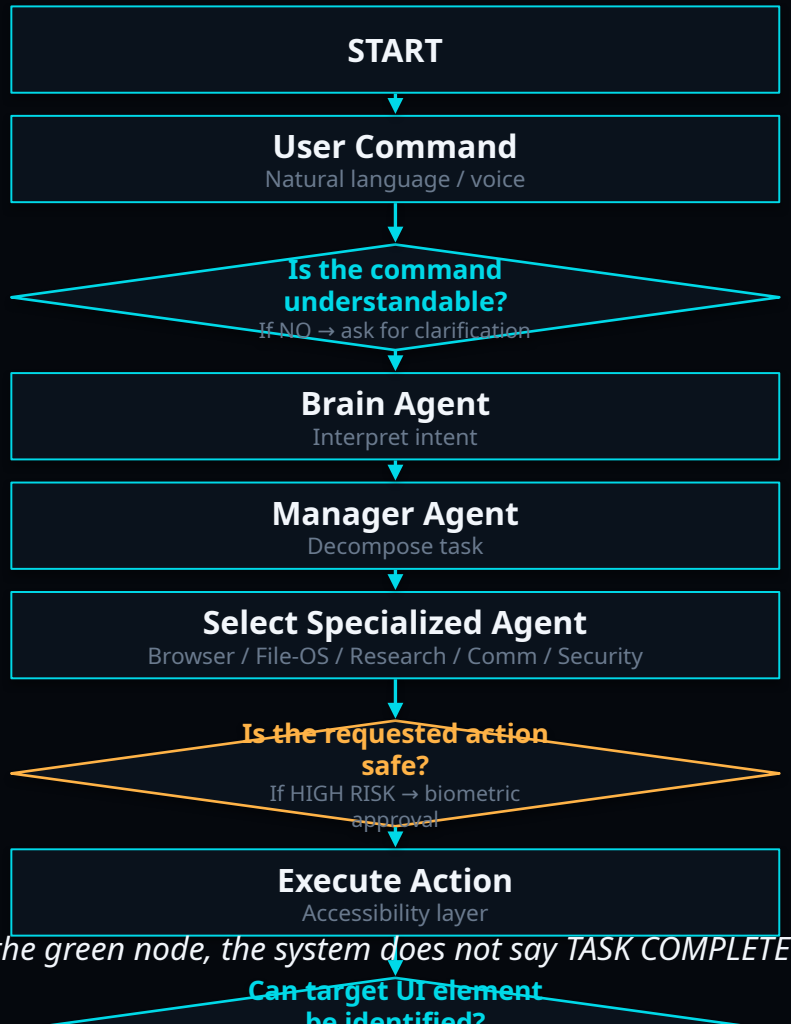
WHAT THIS SLIDE DOES

A.U.R.A. treats every command as a state machine.

Plain stages (cyan) move the task forward.

Diamond decisions branch the task. Two of them — comprehension and risk — can pause or abort.

The amber diamond is the only point where A.U.R.A. asks the user for a physical, not digital, decision.



KEY

□ **STAGE**

◇ **DECISION**

AMBER

physical / security gate

GREEN

task completion

If the flowchart cannot reach the green node, the system does not say TASK COMPLETED.

Layered System Architecture

Five layers, top to bottom. The aesthetic matches the device render.

REQUEST ↓

USER INTERACTION LAYER

Desktop Interface · Natural-language commands, task status, execution feedback

IoT Companion · Raspberry Pi — wake-word mic, fingerprint sensor, projector

ORCHESTRATION LAYER

Brain · Intent interpretation

Manager · Planning + routing + shared context

INTELLIGENCE LAYER

Lightweight · BitNet b1.58 2B

Complex · Quantized 7–8B model · llama.cpp-based inference

EXECUTION LAYER

Accessibility APIs · Direct UI targeting and application interaction

Restricted Execution Env. · Controlled permissions · sandboxed operations

DATA LAYER

Memory · ChromaDB

State · SQLite

Protection · AES-256 encrypted local storage

STATE ↑

User input enters at the top. State (memory, execution logs) persists at the bottom in encrypted storage.

Security by Design

Security is integrated into the architecture, not bolted on at the end.

01

LOCAL-FIRST PROCESSING

AI inference and user context stay on the local machine.

02

ZERO-TRUST INPUT HANDLING



Indirect prompt-injection surface is reduced.

03

RESTRICTED PERMISSIONS

Agents receive only the permissions required for the current task.

04

HUMAN-IN-THE-LOOP



Amber stages = physical pause.

05

ENCRYPTED LOCAL DATA

AES-256 protects sensitive state and memory stores.

06

AUDITABLE EXECUTION

Important decisions and actions are logged for user review.

Honest caveat: this is architecture-level assurance, not formal verification. Future work includes sandboxing audits and prompt-injection testing.

A.U.R.A.

Air-gapped · Unified · Responsive · Agent

A local AI system designed to turn natural-language commands into controlled computer actions.



LOCAL INTELLIGENCE

Reasoning operates locally without cloud AI APIs.

SMART AUTOMATION

Multi-agent architecture decomposes and routes complex tasks.

RESILIENT INTERACTION

Accessibility primary, vision as fallback.

PRIVACY & SECURITY

Sanitization · restricted permissions biometric approval · encrypted storage.

AI that can act on your computer

— while the computer remains under your control.

THANK YOU

Muhammad Hunzala Hasan — 25435 · Muhammad Taha Jabbar — 25421 · Hafsa Ayub — 25588

ADVANCED · UNIFIED · RESPONSIVE · ASSISTANT